

04 Benchmark and Ablation Report

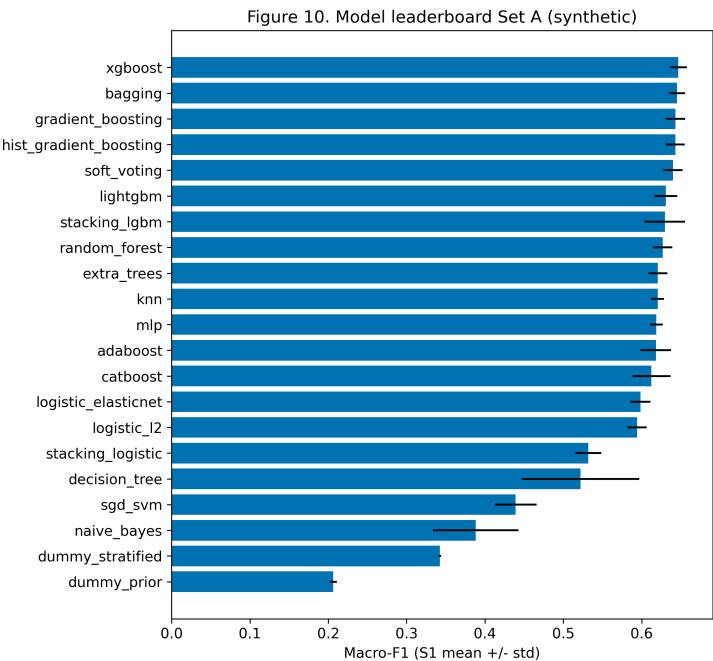
Disclosure: all results use the synthetic fra_synthetic_claims dataset. Scores may reflect generator rules. Real-world performance is unknown. This demo must not drive real claim decisions.

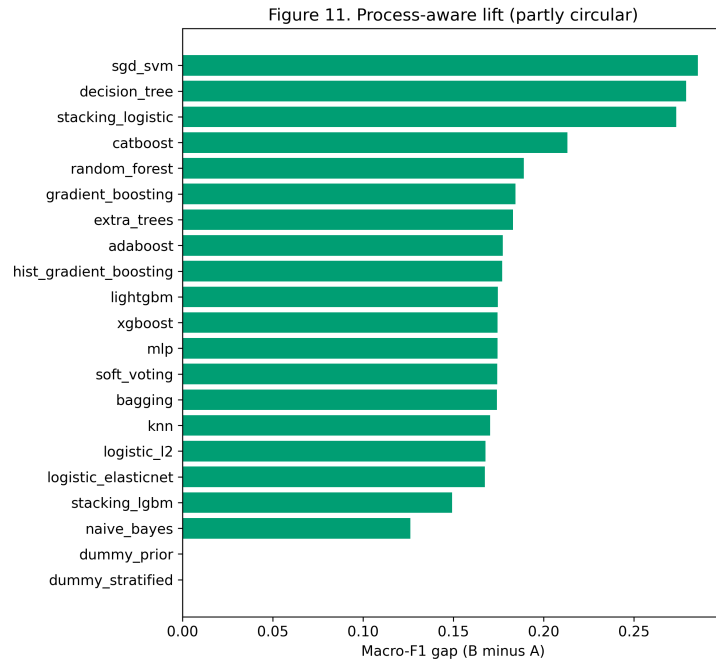
1. Split strategies

S1: StratifiedGroupKFold by district (primary). S2: random stratified (optimistic). S3: temporal train through 2020, test 2021 to 2024. S4: leave-one-state-out. Locked district-held-out test used once per model.

2. Champion

```
{
  "model": "xgboost",
  "feature_set": "A",
  "s1_macro_f1_mean": 0.6468642536953271,
  "s1_macro_f1_std": 0.0109464513248765,
  "s1_log_loss_mean": 0.7040239810943604,
  "s3_macro_f1": 0.5211228499255361,
  "s3_accuracy": 0.5011820330969267,
  "final_macro_f1": 0.5690448456514766,
  "final_accuracy": 0.6757690978223297,
  "final_log_loss": 0.6494698524475098,
  "latency_ms_per_1k": 6.068873142453688,
  "train_seconds": 0.3852621999976691,
  "error": null
}
```





3. Extended tests

McNemar (champion vs runner-up): {"a": "xgboost", "b": "lightgbm", "statistic": 14.822015725518227, "pvalue": 0.00011814815900463112, "n01": 627, "n10": 772}.

Bootstrap 95% CI macro-F1: {"xgboost": {"mean": 0.5693665955517334, "ci_low": 0.556222927577374, "ci_high": 0.5804961345473231}, "lightgbm": {"mean": 0.6289945634991585, "ci_low": 0.6167149618886016, "ci_high": 0.6390508359036629}, "n_boot": 200}

Friedman/Nemenyi: {"friedman_stat": 23.14285714285714, "friedman_p": 0.0007500168565731797, "models": ["lightgbm", "xgboost", "random_forest", "hist_gradient_boosting", "catboost", "soft_voting", "bagging"], "nemenyi_pvalues": {"lightgbm": {"lightgbm": 1.0, "xgboost": 0.053, "random_forest": 0.9972, "hist_gradient_boosting": 0.6757, "catboost": 0.9995, "soft_voting": 0.844, "bagging": 0.2239}, "xgboost": {"lightgbm": 0.053, "xgboost": 1.0, "random_forest": 0.0081, "hist_gradient_boosting": 0.844, "catboost": 0.0134, "soft_voting": 0.6757, "bagging": 0.9972}, "random_forest": {"lightgbm": 0.9972, "xgboost": 0.0081, "random_forest": 1.0, "hist_gradient_boosting": 0.2976, "catboost": 1.0, "soft_voting": 0.4782, "bagging": 0.053}, "hist_gradient_boosting": {"lightgbm": 0.6757, "xgboost": 0.844, "random_forest": 0.2976, "hist_gradient_boosting": 1.0, "catboost": 0.9995, "soft_voting": 0.844, "bagging": 0.2239}, "catboost": {"lightgbm": 0.9995, "xgboost": 0.0134, "random_forest": 1.0, "hist_gradient_boosting": 0.9995, "catboost": 1.0, "soft_voting": 0.844, "bagging": 0.2239}, "soft_voting": {"lightgbm": 0.844, "xgboost": 0.6757, "random_forest": 0.4782, "hist_gradient_boosting": 0.844, "catboost": 0.844, "soft_voting": 1.0, "bagging": 0.2239}, "bagging": {"lightgbm": 0.2239, "xgboost": 0.9972, "random_forest": 0.053, "hist_gradient_boosting": 0.2239, "catboost": 0.2239, "soft_voting": 0.2239, "bagging": 1.0}}}

Figure 14. S1 vs S3 (synthetic)

